Analyse — Reservoir Decision Gate Tracking

Select a reservoir to compare Business Decisions across decision gates. Track how input parameters, volumes, economics, and risks evolve from early screening to concept selection.

1 — Select Reservoir

*

Search

RES-001 — dev:master-data--Reservoir:e06b1a03-b397-46e6-8e7f-cf8b40ba6f30:1
RES-001 — dev:master-data--Reservoir:93f292e5-d9bb-47ce-9d52-81ba5034e633:1
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Drogon — dev:master-data--Reservoir:a88bdb2d-6703-4054-bfe4-62cf80d2bd1e:1
Drogon — dev:master-data--Reservoir:5b8dc759-4be9-493a-9bd3-eb0e55fd078e:1

Loaded 2 decision gate(s).

Load Decision Gates

Drogon

dev:master-data--Reservoir:5b8dc759-4be9-493a-9bd3-eb0e55fd078e:1

DG1	→	DG2
2026-02-28		2026-06-30
Approved		Pending

VOLUMES (MSM³)

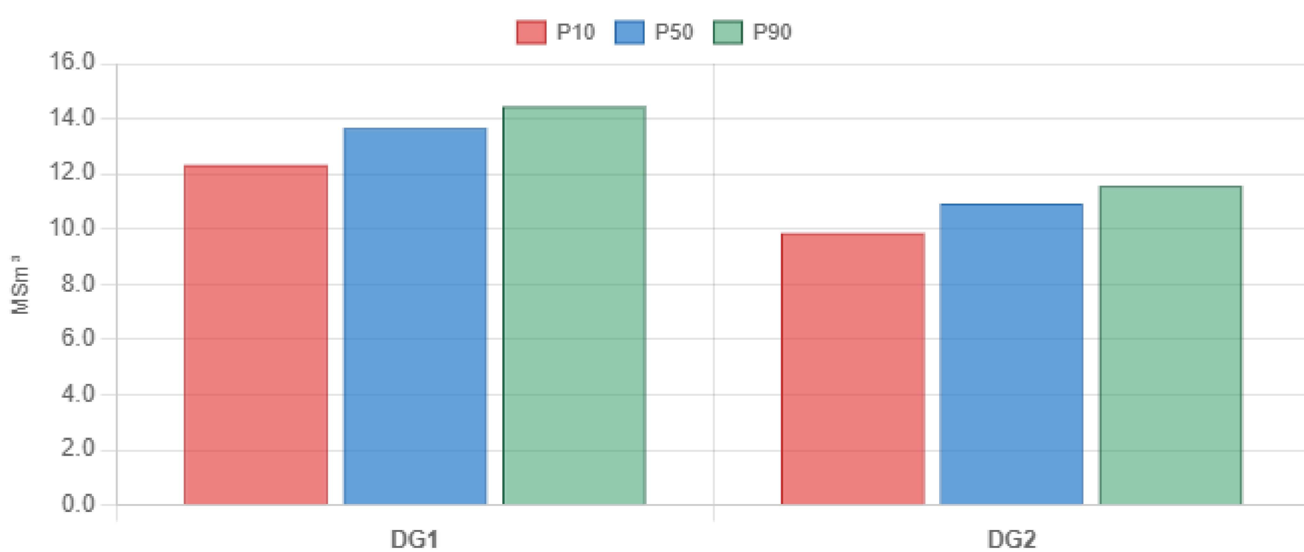
METRIC	DG1	DG2	Δ (LAST STEP)
STOIIP P90	14.4	11.6	-2.89 (-20%)
STOIIP P50	13.7	10.9	-2.73 (-20%)
STOIIP P10	12.3	9.85	-2.46 (-20%)
Recoverable P90	4.33	3.47	-0.866 (-20%)
Recoverable P50	4.71	3.77	-0.943 (-20%)

METRIC	DG1	DG2	Δ (LAST STEP)
Recoverable P10	4.80	3.84	-0.961 (-20%)
Rec. Factor P90 (%)	30	30	0 (0%)
Rec. Factor P50 (%)	34.5	34.5	0 (0%)
Rec. Factor P10 (%)	39	39	0 (0%)

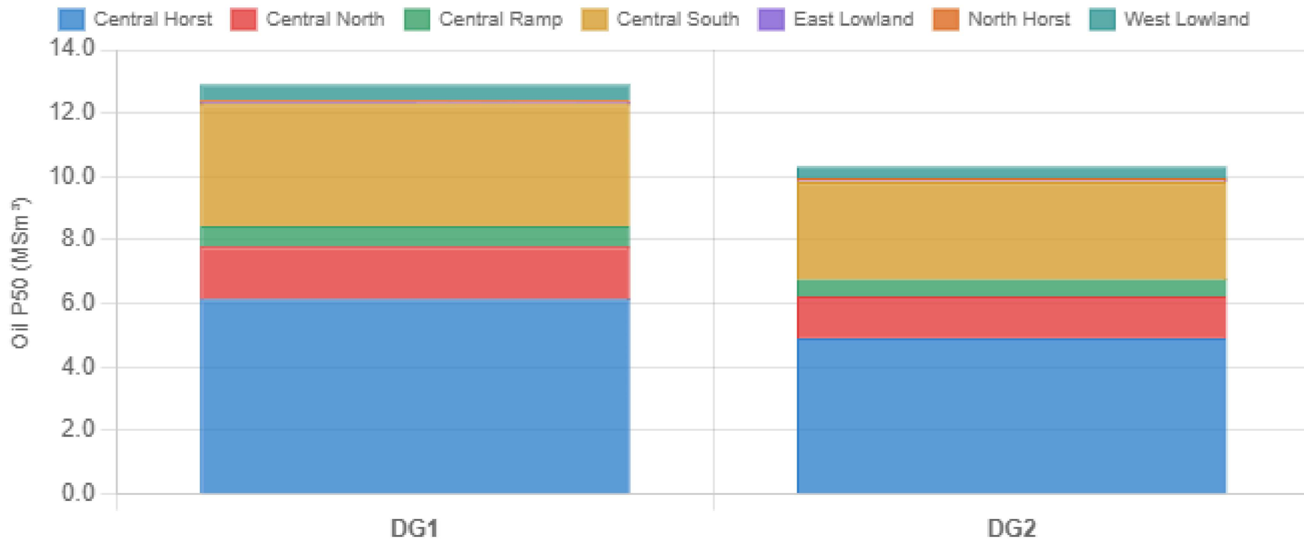
STOIIP BY SEGMENT — P50 (MSM³)

SEGMENT	DG1	DG2	Δ LAST
TOTAL	13.7	10.9	-2.73 (-20%)
Central Horst	6.14	4.91	-1.23 (-20%)
Central North	1.65	1.32	-0.330 (-20%)
Central Ramp	0.658	0.527	-0.132 (-20%)
Central South	3.84	3.07	-0.767 (-20%)
East Lowland	0.031	0.025	-0.006 (-20%)
North Horst	0.110	0.088	-0.022 (-20%)
West Lowland	0.466	0.373	-0.093 (-20%)

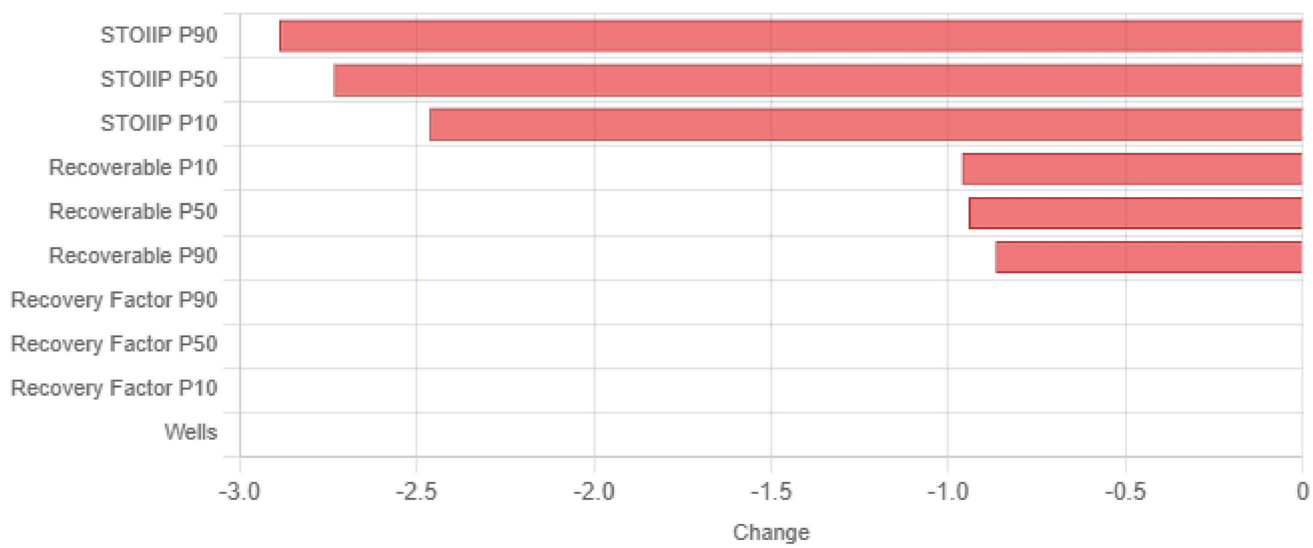
STOIIP UNCERTAINTY RANGE BY GATE



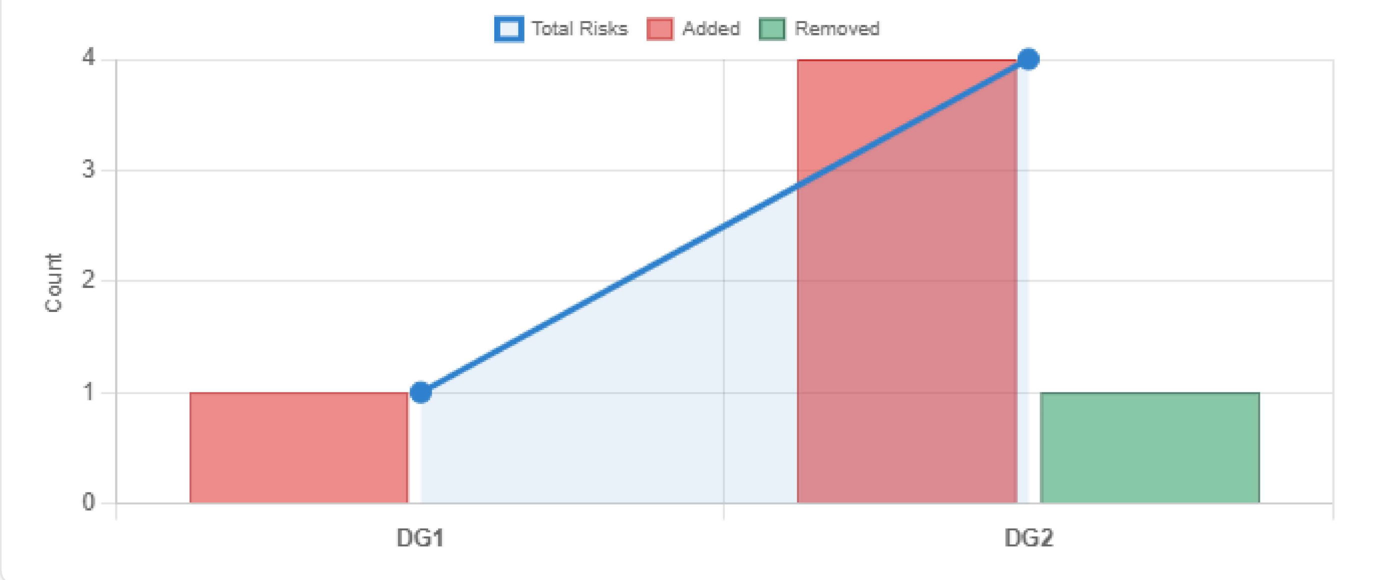
OIL IN PLACE BY SEGMENT



DELTA TORNADO — LAST GATE CHANGE



RISK EVOLUTION



ECONOMICS

METRIC	DG1	DG2	Δ (LAST STEP)
NPV (MUSD)	—	520	*new*
CAPEX (MNOK)	—	8,500	*new*
OPEX (MNOK/yr)	—	420	*new*
IRR (%)	—	17	*new*
Breakeven ($/bbl)	—	42	*new*

DEVELOPMENT CONCEPT

PARAMETER	DG1	DG2
Wells	12	12
Template Slots	16	10
Drilling Centres	—	2
Host Facility	—	Drogon FPSO (converted)
Distance to Host (km)	—	8
Water Depth (m)	108	108
Formation	Valysar	Valysar
Field Area	Drogon	Drogon
Target Start-up	2028-H1	2028-H1

PARAMETER	DG1	DG2
Injection Strategy	—	Water injection for pressure support (4 injectors planned Phase 2)
Summary	Subsea development with tie-back to existing host facility. Valysar formation at ~1700 m TVD MSL in the Drogon area, Norwegian North Sea.	Subsea development with 2×4-slot templates (8 slots + 2 contingent), tie-back to converted FPSO via dual 10" production flowlines and 6" gas-lift line. Distance to FPSO ~8 km. Water depth 108 m. Valysar formation at ~1700 m TVD MSL.

RESERVOIR PROPERTIES (GEOLABELSET)

PROPERTY	DG1	DG2	Δ LAST
FormationVolumeFactor	1.12	1.12	0 (0%)
GasOilRatio	85	85	0 (0%)
NetToGross	0.8500	0.8500	0 (0%)
OilWaterContact	1,710	1,710	0 (0%)
Permeability	450	450	0 (0%)
Porosity (Channel)	0.2800	0.2800	0 (0%)
Porosity (Crevasse)	0.2100	0.2100	0 (0%)
Porosity (Floodplain)	0.1000	0.1000	0 (0%)
Pressure	170	170	0 (0%)
Temperature	72	72	0 (0%)
Viscosity	1.20	1.20	0 (0%)

RISK EVOLUTION

GATE	RISKS	CHANGES
DG1	Drogon — Porosity and cementation uncertainty	Baseline
DG2	Drogon DG2 — Porosity and cementation uncertainty Drogon DG2 — Fault transmissibility and reservoir compartmentalization	+ Drogon DG2 — Fault transmissibility and reservoir compartmentalization + Drogon DG2 — HSE and environmental impact

GATE	RISKS	CHANGES
	Drogon DG2 — HSE and environmental impact Drogon DG2 — Schedule risk (FPSO and long-lead equipment)	+ Drogon DG2 — Porosity and cementation uncertainty + Drogon DG2 — Schedule risk (FPSO and long-lead equipment) ~~– Drogon — Porosity and cementation uncertainty~~

KEY UNCERTAINTIES

GATE	UNCERTAINTIES
DG1	**[High]** Facies-dependent porosity — Porosity varies 0.10–0.28 across facies; Channel-dominant segments (CentralHorst) have higher confidence than Floodplain-rich segments (NorthSea, EastLobe). **[Medium]** OWC depth and aquifer support — OWC modelled at 1710 m TVDSS with ±15 m uncertainty; deeper OWC would increase STOIIP but also water production risk. **[Medium]** Cementation and diagenesis — Deeper segments show evidence of diagenetic cementation reducing effective porosity and permeability, particularly in Crevasse facies.
DG2	**[Medium]** Facies-dependent porosity — DG2 appraisal core data have narrowed porosity ranges (revised ×0.8 from DG1): Channel 0.21–0.24, Crevasse 0.15–0.18, Floodplain 0.06–0.10. Impact downgraded from High (DG1) to Medium based on 50-realisation FMU results. STOIIP reduced by ~20% vs DG1. **[High]** Fault transmissibility — Production testing confirmed partial communication across CentralHorst–CentralSouth fault. NorthHorst and EastLobe remain untested. 30% probability of needing 2–4 infill wells (80–120 MUSD incremental CAPEX). **[Medium]** OWC depth and aquifer support — OWC refined to 1710 ± 10 m TVDSS (from ±15 m at DG1). Aquifer modelling suggests moderate pressure support, sufficient for primary depletion; water injection reserved for Phase 2. **[High]** Recovery factor uncertainty — Dynamic simulation (50 realisations, revised porosity) gives RF range 28–37%. P50 = 32.5%. Absolute recoverable reduced by ~20% vs DG1 (14.8 vs 18.5 MSm^3). Main driver is sweep efficiency in NorthHorst and EastLobe where fault baffles may create unswept compartments. ICD completions and Phase 2 water injection are key mitigations. **[Medium]** FPSO water treatment capacity — Water cut expected to reach 50% by Year 5 and 70% by Year 10. FPSO design includes 5,000 m^3/d treatment; expansion module to 8,000 m^3/d available for Year 7+. Uncertainty in early water breakthrough timing from EastLobe aquifer.

Synthesis — Cross-Gate Change Analysis

STOIIP P50 decreased by 20% (-2.73 MSm^3) from DG1 to DG2.

Recoverable oil P50 decreased by 20% from DG1 to DG2.

NPV first quantified at DG2: $520M.

4 new risk(s) identified, 1 risk(s) mitigated/removed between DG1 and DG2.

Mitigation focus for next gate: address new risks and validate subsurface model assumptions to reduce volume uncertainty.